Rave Generation

PRESENTS

SONIC SURGE 2

USER MANUAL

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1. Introduction

Sonic Surge 2 is an advanced multiband dynamics processor that combines optical compression modeling with modern enhancement features for contemporary music production. Building on the success of the original Sonic Surge, this second generation brings 3-band processing with precise control over dynamics, harmonics, and spatial characteristics.

At its core, Sonic Surge 2 features stereo-linked optical compression that can operate in either Multiband or Fullband mode, inspired by classic compressors used in mastering studios. Beyond traditional compression, the plugin introduces modern enhancements including per-band tone curves, transient shaping, and low-frequency relaxation for transparent bass control.

Whether you're adding punch to EDM drops, controlling dynamics in complex mixes, enhancing individual instruments, or mastering full productions, Sonic Surge 2 provides the character of analog compression with the precision and flexibility required for modern production.

2. Key Features

- **Dual-mode operation:** Switch between Multiband (3-band independent) and Fullband (unified) compression modes
- **3-band optical compression modeling:** Stereo-linked operation with per-channel optical cell variation for authentic stereo punch on transients
- **Selectable crossover slopes:** 6 dB/oct (authentic Tube-Tech SMC 2B style) or 24 dB/oct Linkwitz-Riley. Optional x4 multiplier extends low crossover range in 6 dB mode
- **Per-band tone curves:** Inflator-style waveshaping (positive values) or dynamic expansion (negative values)
- **Per-band gain controls:** Independent band gains active in both compression modes
- **Per-band mute switches:** Solo or remove individual frequency bands in any mode
- **Flexible compression:** Full control over threshold, ratio, attack, and release per band
- **SC HPF:** High-pass filtering of the compression sidechain (0–1000 Hz)
- **Mono Maker:** M/S processing to collapse low frequencies to mono (0–1000 Hz)
- **Detection modes:** Stereo (shared linked detection), L/R (independent per-channel detection), and M/S (Mid/Side encode with independent detection)
- **M/S Balance:** Stereo width control (±1.5 dB) operating in the Mid/Side domain regardless of detection mode
- **Dynamic Depth:** Transient enhancement with soft limiting (0–100%)
- **Harmonic Drive:** Tape-like saturation modeling
- **Clip 0dB:** Dual-stage soft/hard clipping at 0 dB
- **Listen Mode:** Monitor only the processed portion of the signal
- **Tone Mode:** Place tone curves before (Pre-Comp) or after (Post-Comp) compression
- **Visual feedback:** Real-time gain reduction meters and LED indicators per band

3. Quick Start

1. Insert Sonic Surge 2 on your desired track, bus, or master channel.
2. Enable the effect using the **Effect In** switch (set to “on”).
3. Set input level with **INPUT** for optimal signal level into processing.
4. Choose mode: Select between **Multiband** or **Fullband** in Comp Mode.
5. Set crossover points: Adjust **X-OVER LOW** (60–300 Hz) and **X-OVER HIGH** (1200–6000 Hz).
6. Enable compression (Comp Enable) and adjust **THRESHOLD** for each band (0 dB = off).
7. Adjust per-band **GAIN dB** controls to balance frequency content.
8. Add enhancement: Try **DYNAMIC DEPTH** for punch, **HARMONIC DRIVE** for warmth.
9. Fine-tune per-band **TONE** controls for character.
10. Set output level with **OUTPUT** and enable **Clip 0dB** if maximizing loudness.

4. User Interface

The interface is organized in four rows: three band rows (High, Mid, Low from top to bottom) and a Master row at the bottom. An FX column sits on the right. Each band row follows the same layout from left to right, making it easy to learn once and apply everywhere.

4.1 Input & Crossover

These controls sit on the left side of the interface, shared across all bands.

Control	Range	Description
INPUT	-12 to +12 dB	Input level control before all processing.
X-OVER LOW	60-300 Hz	Low/Mid crossover frequency. Active in both modes.
X-OVER HIGH	1200-6000 Hz	Mid/High crossover frequency. Active in both modes.
X-Over Slope	6 dB / 24 dB	Crossover filter slope. 6 dB/oct (authentic Tube-Tech SMC 2B) or 24 dB/oct (Linkwitz-Riley, default).
X-Over x4	off / on	Multiplies low crossover frequency by 4 (only visible in 6 dB slope mode). Extends range to 240-1200 Hz.

4.2 Band Controls (Low / Mid / High)

All three frequency bands share the same set of controls. Each band row runs left to right: GR meter, Threshold, Ratio, Attack, Release, Gain dB, Tone, Mute.

Compression

In Multiband mode each band compresses independently. In Fullband mode only the High band parameters are used (they control the unified compressor). A threshold of 0 dB means compression is off for that band.

Control	Range	Default	Description
THRESHOLD	-60 to 0 dB	0 dB	Compression threshold. 0 dB = off.
RATIO	1.5:1 to 10:1	2.0:1	Compression ratio.
ATTACK	0.5-200 ms	30 ms	Attack time.
RELEASE	50-2000 ms	200 ms	Release time.

Gain, Tone & Mute

Control	Range	Description
GAIN dB	-12 to +12 dB	Per-band gain applied after compression. Active in both modes.
TONE	-50 to +50 %	Per-band tone curve. Positive = inflator-style waveshaping. Negative = dynamic expansion. Active in both modes.
Mute	off / on	Mute the band with a smooth 5 ms transition. Active in both modes.

GR Meter

Each band row has a vertical gain reduction meter to the left of the compression knobs. The meter fills downward with a green-to-red color gradient. In Fullband mode only the High band meter is active.

4.3 FX Section

The FX column sits on the right side of the interface, stacked vertically alongside the three band rows.

Control	Range	Description
DYNAMIC DEPTH	0–100 %	Transient enhancement amount with soft limiting. Applied per band.
HARMONIC DRIVE	0–100 %	Tape-style saturation intensity. Applied per band.
DRY/WET	0–100 %	Global wet/dry mix of all processing.

4.4 Master Section

The bottom row contains global controls that affect the entire signal path.

Control	Range	Description
SC HPF	0–1000 Hz	High-pass filter on compression sidechain (0 = off). Uses cascaded 24 dB/oct filters. Active in all compression and detection modes.
Comp Enable	off / on	Global compression bypass.
Comp Mode	Multiband / Fullband	Switch between 3-band independent and unified fullband compression.
X-Over Slope	6 dB / 24 dB	Crossover filter slope selector (also listed in section 4.1).
Tone Mode	Post-Comp / Pre-Comp	Position tone curves before or after compression.
Detection	Stereo / L/R / M/S	Compression detection mode. Stereo = shared linked detection. L/R = independent per-channel detection. M/S = Mid/Side encode with independent detection per channel.
M/S BALANCE	±1.5 dB	Stereo width control operating in Mid/Side domain. Positive = boost Side (wider). Negative = boost Mid (narrower). Active regardless of Detection mode.
Listen Mode	off / on	Monitor only the processed (wet) portion of the signal.
MONO MAKER	0–1000 Hz	Frequency below which the stereo signal is summed to mono (0 = off).
OUTPUT	-12 to +12 dB	Final output level control.
Clip 0dB	off / on	Dual-stage soft/hard clipping at 0 dB.
Effect In	off / on	Master effect bypass with smooth 5 ms transition.

5. Signal Flow & Processing

Complete Signal Path

Multiband Mode: Input → INPUT gain → 3-Band Crossover → [Tone Curve if Pre-Comp] → Per-Band Compression → Per-Band GAIN dB → Dynamic Depth → Harmonic Drive → [Tone Curve if Post-Comp] → DRY/WET Mix → OUTPUT gain → Clip 0dB → Band Mute → Band Sum → Mono Maker → Final Clip → Output

Fullband Mode: Input → INPUT gain → 3-Band Crossover → [Tone Curve if Pre-Comp] → Fullband Compression (applied proportionally) → Per-Band GAIN dB → Dynamic Depth → Harmonic Drive → [Tone Curve if Post-Comp] → DRY/WET Mix → OUTPUT gain → Clip 0dB → Band Mute → Band Sum → Mono Maker → Final Clip → Output

M/S Detection Mode: When Detection is set to M/S, the signal is encoded to Mid/Side after INPUT gain and before the crossover. All processing runs on the Mid and Side signals independently. After band summing, M/S BALANCE is applied, then the signal is decoded back to L/R before Mono Maker and final output.

L/R Detection Mode: When Detection is set to L/R, the signal path is the same as Stereo mode but each channel has its own independent compressor detecting and compressing separately. M/S BALANCE is still applied in the Mid/Side domain at the output stage.

3-Band TPT Crossover

The crossover is active in both Multiband and Fullband modes. It uses a serial topology: the input first splits into Low and (Mid+High), then (Mid+High) splits into Mid and High. Two slope options are available:

- **6 dB/oct:** One-pole TPT lowpass with complement. Authentic Tube-Tech SMC 2B style with perfect unity sum and zero phase dips.
- **24 dB/oct:** 4th-order Linkwitz-Riley SVF. Sharp band separation with allpass sum (default).

Per-Band Processing

Each frequency band passes through the same processing chain:

11. Tone Curve (if Pre-Comp mode is selected)
12. Compression (independent per band in Multiband; proportional in Fullband)
13. GAIN dB (post-compression band gain)
14. Dynamic Depth (transient shaping)
15. Harmonic Drive (saturation)
16. Tone Curve (if Post-Comp mode is selected)
17. DRY/WET mix (blend with original band signal)
18. OUTPUT gain
19. Clip 0dB (soft clip per band if enabled)
20. Band Mute

Fullband Mode Specifics

In Fullband mode compression is applied to the summed signal using the High band's compression parameters. The resulting gain reduction is then distributed proportionally to each band, maintaining spectral balance while applying unified dynamics. All other band processing (GAIN dB, Tone, Mute) remains fully active.

Stereo Processing & Detection Modes

The Detection switch selects how the compressor's sidechain operates across stereo channels. Three modes are available:

Stereo (default): Shared detection with linked gain reduction. One detector per band averages the L+R signal and applies identical compression to both channels. This preserves the stereo image and produces the tightest, loudest result. Per-channel optical cell variation still creates subtle stereo widening on transients.

L/R: Independent detection per channel. The left and right channels each have their own compressor instance detecting and compressing independently. Peaks on one side are caught without affecting the other, producing a more even and controlled result with slightly increased stereo depth. Uses the same knob settings for both channels.

M/S: Mid/Side encode with independent detection. The stereo signal is encoded to Mid (center content: kick, bass, vocal) and Side (stereo content: reverb, panned elements) before entering the crossover and compression chain, then decoded back to L/R at the output. Each channel compresses independently. Because the Mid signal is typically louder, it compresses harder while the Side signal breathes freely, creating a wider, more spacious sound.

The M/S BALANCE knob (±1.5 dB) operates in the Mid/Side domain regardless of Detection mode. Turning it positive boosts the Side signal (wider stereo image), while negative boosts the Mid signal (narrower, more focused center). At 0 dB the stereo balance is unchanged. The GR meters show the maximum gain reduction across both channels in L/R and M/S modes.

SC HPF (Sidechain High-Pass Filter)

When SC HPF is set above 0 Hz the compression detector signal is high-pass filtered at the specified frequency using cascaded 24 dB/oct filters. This prevents low-frequency content from triggering gain reduction while still being processed by other effects (Tone, GAIN dB, Harmonic Drive), resulting in more transparent bass handling. Active in all compression and detection modes.

Mono Maker

When MONO MAKER is set above 0 Hz an M/S high-pass filter is applied to the side signal at the specified frequency. Frequencies below this point are collapsed to mono for a tighter low end, while higher frequencies retain their full stereo width.

6. Operating Modes

Multiband Mode

The default mode offering independent compression of three frequency bands:

- Each band has its own Threshold, Ratio, Attack, and Release
- Full crossover network active for band separation
- All three GR meters show independent activity
- Best for surgical compression and problem frequency management

Fullband Mode

Unified compression mode with multiband tone shaping:

- Single compression stage using the High band parameters applied to the full spectrum
- Crossover remains active for band-specific tone shaping and gain control
- All GAIN dB, Tone, and Mute controls remain fully active per band
- Compression is applied to the sum, then scaled proportionally across bands
- Only the High band GR meter shows activity
- Best for cohesive “glue” compression while maintaining tonal control

Important: In Fullband mode the Low and Mid compression parameters are ignored. Only the High band’s Threshold, Ratio, Attack, and Release control the unified compressor.

7. Tips & Tricks

The “Sonic Surge” Sound

21. Start with Low GAIN dB at +3 dB, Mid GAIN dB at +1 dB for warmth.
22. Set DYNAMIC DEPTH to 25–40 % for punch.
23. Add 15–30 % HARMONIC DRIVE for analog character.
24. Enable Clip 0dB for transparent loudness.
25. Use Multiband for precision, Fullband for glue, Pre-Comp for vintage character.

Electronic Music Production

- **Kick Enhancement:** Low GAIN dB +4 dB, Low Attack 30 ms, Low Release 200 ms
- **Drop Impact:** DYNAMIC DEPTH 60–80 %, all bands compressed
- **Wide Mix:** High GAIN dB +2 dB, HARMONIC DRIVE 20 %
- **Sub Control:** MONO MAKER at 120 Hz for tight low end
- **Fullband Glue:** Switch to Fullband mode, 2:1 ratio, slow attack

Vocal Processing

- **Presence:** Mid GAIN dB +2–4 dB, Mid Tone +20 %
- **Air:** High GAIN dB +1–2 dB, High Tone +30 %
- **Warmth:** Low GAIN dB +1 dB at 150 Hz crossover
- **Control:** Use Fullband mode for natural compression with tonal shaping

Mastering Applications

- **Transparent Enhancement:** DRY/WET at 40–60 %
- **Multiband Control:** Multiband mode with gentle ratios (2:1 to 3:1)
- **SC HPF:** SC HPF at 80–120 Hz for open bass
- **Tone Curves:** Subtle ±15 % for character without artifacts
- **Final Glue:** Switch to Fullband for cohesion, use GAIN dB for balance

Tone Mode

Post-Comp (default): Clean, predictable compression followed by tonal enhancement. Tone curves add harmonics to already-controlled dynamics. Best for mastering and transparent processing.

Pre-Comp: Tone curves feed harmonics into the compressor detection circuit, creating more colored, vintage-style compression behavior. The compressor “hears” and reacts to the added harmonics. Expansion curves (negative values) create gate-like effects.

Creative uses:

- **Vintage Warmth:** Pre-Comp with +20-30 % curves for analog-style compression
- **Dynamic Enhancement:** Pre-Comp with negative curves for expansion before compression
- **Punchy Drums:** Pre-Comp emphasizes transients that trigger the compressor

Mode Selection Guidelines

Use Multiband Mode when:

- You need independent compression per frequency band
- Dealing with specific problem frequencies
- Processing unbalanced material
- You need to compress only certain frequency ranges

Use Fullband Mode when:

- You want cohesive "glue" compression across the spectrum
- Need unified compression with multiband tonal shaping
- Seeking classic compressor behavior with modern flexibility

Compression Strategies

- **Transparent:** Thresholds at -5 to -10 dB, low ratios (1.5:1 to 2:1)
- **Punchy:** Medium attack (10-30 ms), medium release (100-300 ms)
- **Pumping:** Fast release times (50-100 ms), higher ratios (4:1+)
- **Glue:** Fullband mode, slow attack (30-50 ms), 2:1 to 4:1 ratios

Band Isolation Techniques

- Use Mute switches to solo bands for precise adjustment
- Listen Mode reveals exactly what's being added to the signal
- Adjust crossovers while listening to isolated bands
- Works in both Multiband and Fullband modes

8. Installation

System Requirements

- **Operating System:** macOS 10.13+ or Windows 10+
- **Processor:** Intel Core i5 or equivalent (Apple Silicon native support)
- **RAM:** 4 GB minimum (8 GB recommended)
- **Plugin Formats:** VST3, AU (macOS)
- **DAW:** Any VST3/AU compatible host

Support

For additional support, preset packs, and tutorials, visit **ravegeneration.io**

Sonic Surge 2 by Rave Generation - Modern multiband dynamics processing with the character of classic optical compression. Experience precise frequency control with the musical response that made analog gear legendary.

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